

Development of an ESP32-S3 Hydroponic Nutrient Control System with Flow-Based Safety Interlock

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Received: July 22nd, 2026. Revised: August 31st, 2026. Accepted: September 3rd, 2026

Available online: September 9th, 2026. Published: September 9th, 2026.

Abstract— Hydroponic cultivation requires precise control of nutrient concentration and pH, while manual monitoring and adjustment may result in inconsistent nutrient conditions and increased labor requirements. This study aimed to develop and evaluate an ESP32-S3-based automated hydroponic nutrient management system for water filling, nutrient dosing, pH adjustment, circulation verification, and scheduled nutrient solution replacement. A quantitative Research and Development (R&D) approach was employed to design, implement, and evaluate the system, integrating EC/TDS, pH, ultrasonic, water flow, temperature and humidity sensors, a real-time clock, peristaltic pumps, solenoid valves, a circulation pump, and a 20×4 I2C LCD. Experimental testing included sensor calibration, actuator functional testing, flow-based safety interlock testing, and integrated system operation. The results showed that the pH sensor achieved a maximum measurement error of 0.43%, while the EC sensor produced errors of 5.16–13.23% at 1.413 mS/cm and 2.17–2.95% at 12.88 mS/cm. The ultrasonic sensor produced errors of 0% in four measurements and 5% in one measurement. The flow sensor successfully distinguished flow and no-flow conditions and prevented nutrient and pH dosing when circulation was absent. The peristaltic pumps achieved an average flow rate of 0.908 mL/s, and the integrated system completed the tested automation sequence. However, testing was limited by the number of calibration points and short-duration functional evaluation, with no assessment of long-term nutrient stability, repeated dosing accuracy, or plant-growth performance.

Keywords: ESP32-S3, hydroponic automation, nutrient control, flow sensor, safety interlock.

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I. INTRODUCTION

Technological advancements in agriculture continue to accelerate in response to increasing food demand and the limited availability of agricultural land, particularly in urban areas. Hydroponic cultivation has become one of the most promising solutions because it enables soilless farming using nutrient solutions as the primary source of plant nutrition. Compared with conventional cultivation, hydroponic systems provide advantages such as efficient water use, easier environmental control, and faster plant growth when nutrient parameters are maintained within optimal ranges (Arivin Rivaie et al., 2021; Hariono, Putra, & Hasbullah, 2021).

The quality of the nutrient solution is a critical factor determining the success of hydroponic cultivation. The most important parameters are Total Dissolved Solids (TDS), which represents nutrient concentration, and pH, which determines nutrient availability for plant absorption (Marwah, 2022). For hydroponic vegetables, TDS is generally maintained between 500–1400 ppm depending on plant species and growth stage, while the pH should remain within 5.5–6.5. Deviations from these values reduce nutrient uptake efficiency and may cause nutrient deficiencies, growth disorders, and root damage,

highlighting the need for continuous monitoring and control (Food and Agriculture Organization (FAO), 2023; Homepage & Fahlefi, 2025).

Besides controlling TDS and pH, periodic replacement of the nutrient solution is essential to prevent mineral salt accumulation, nutrient imbalance, and deterioration of water quality that can inhibit plant growth (Soufi, Roosta, & Hamidpour, 2023). However, nutrient management is still commonly performed manually through periodic measurements and manual nutrient or pH adjustment. This process is labor-intensive, susceptible to human error, and often fails to respond quickly to changes caused by plant nutrient uptake and water evaporation, resulting in unstable nutrient conditions.

Recent developments in microcontroller technology have encouraged the implementation of automated hydroponic systems using platforms such as the ESP32. These systems commonly integrate TDS and pH sensors with peristaltic pumps for automatic nutrient dosing, while additional temperature, humidity, and water-level sensors improve monitoring accuracy and overall system stability (Linelson & Rinanda Saputri, 2025). Nevertheless, most existing systems focus primarily on nutrient control without considering the reliability of nutrient solution circulation.

When TDS and pH sensors are installed in the circulation line, accurate measurements depend on continuous solution flow. If the circulation pump fails, stagnant sensor readings may cause the control system to continue dispensing nutrients, leading to nutrient wastage and overdosing. Therefore, a flow-based safety mechanism is required to ensure that sensor measurements and nutrient dosing are executed only when proper circulation is detected. In addition, this research was motivated by the need for practical educational media capable of integrating sensors, microcontrollers, and actuators into a real-world application, enabling students to better understand automation technology through project-based learning.

To address these challenges, this study proposes an ESP32-based hydroponic nutrient automation system equipped with TDS, pH, temperature and humidity, ultrasonic, and flow sensors. The flow sensor functions as a safety interlock that verifies solution circulation before measurements and nutrient dosing are performed, while the system also incorporates automatic nutrient replacement every two weeks to maintain nutrient quality throughout the cultivation period. Furthermore, the proposed system supports Sustainable Development Goal (SDG) 9 by promoting technological innovation, efficient resource utilization, and sustainable agricultural production, consistent with FAO recommendations for transforming food systems (FAO, 2022; The-Sustainable Development Goals, 2023).

A review of previous studies was conducted to identify research gaps and compare the proposed system with existing hydroponic automation technologies. The first reviewed study presented an IoT-based EC and pH control system for NFT hydroponics that automatically regulated nutrient solutions using EC, pH, and water-level sensors integrated with a microcontroller and remote monitoring, successfully improving nutrient stability and management efficiency (Jain & Kaur, 2024). The second study developed an automated nutrient dosing system for hydroponic strawberries using EC, pH, temperature, and humidity sensors with multiple pumps, demonstrating improved nutrient delivery and reduced manual intervention (F. Putra & Nasuha, 2023).

The third reviewed study introduced an ESP32-based smart hydroponic monitoring system integrating TDS, pH, temperature, and water-level sensors for automatic monitoring and nutrient control. The results demonstrated that combining ESP32 with multiple sensors significantly enhanced automation performance and minimized the need for manual supervision (Sneineh & Shabaneh, 2023). These

previous studies provide the foundation for developing a more comprehensive hydroponic automation system with additional safety features.

The proposed system consists of four integrated modules: sensing, processing, actuation, and display. The sensing module includes TDS, pH, temperature–humidity, ultrasonic, and flow sensors, while the processing module utilizes an ESP32 microcontroller to analyze sensor data (Akbar, Saifullah, & Umami, 2026). The actuator module comprises four peristaltic pumps, two solenoid valves, and a circulation pump controlled through relays (Alimussadad, Lestari, & Utomo, 2025). System operation begins with automatic water filling based on ultrasonic sensing, followed by circulation verification through the flow sensor, TDS and pH measurement, automatic nutrient or pH adjustment, and real-time monitoring using a 20×4 I2C LCD. Finally, a comparison table is presented to evaluate the proposed system against previous studies and highlight its unique contributions and advantages.

II. METHOD

This study employed a quantitative Research and Development (R&D) approach to design and implement an automated hydroponic nutrient management system (Bennett, Borg, & Gall, 1984). The developed system integrates sensors and actuators to automatically regulate water filling, nutrient mixing, pH adjustment, daily monitoring, and periodic water replacement (Herman, Adidrana, Surantha, & Suharjito, 2019). The system operates sequentially according to the flow illustrated in Fig. 1. Initially, when the nutrient tank is empty, an ultrasonic sensor detects the water level. If the water level is below the predefined maximum threshold, the inlet solenoid valve is activated to fill the tank with water (Jain & Kaur, 2024). Once the maximum level is reached, the solenoid valve closes and the circulation pump is turned on to ensure homogeneous mixing of the nutrient solution.

A flow sensor installed in the circulation line serves as a safety mechanism to verify that water flows properly before the nutrient and pH control processes are executed. If no flow is detected, the nutrient pumps and pH adjustment pumps remain inactive to prevent incorrect solution dosing and potential damage to the system. After stable circulation is confirmed, the system measures the Total Dissolved Solids (TDS) value to determine the nutrient concentration. If the measured TDS value is below the predetermined setpoint, nutrient pumps A and B are activated until the concentration approaches the desired range (Yulianto, Nugraha, & Rahami, 2024). Subsequently, the solution continues

circulating for a short period to achieve uniform mixing.

The pH sensor then measures the acidity level of the nutrient solution (Putra, Heriansyah, Cahyadi, Anggriani, & Jaya, 2024). When the pH value is outside the desired range, the system automatically activates the pH-up or pH-down pump until the target pH is attained (Mabitazan & Mabitazan, 2021). During the cultivation period, the system continuously monitors TDS, pH, water level, and flow conditions. If nutrient concentration increases due to water evaporation, the system automatically adds water to restore the solution to the specified conditions. At the end of a two-week cultivation cycle, the system performs an automatic water replacement process by draining the existing nutrient solution, refilling the tank with clean water, and restarting the nutrient preparation cycle from the beginning. This automated control strategy enables continuous monitoring and maintenance of hydroponic nutrient conditions with minimal human intervention.

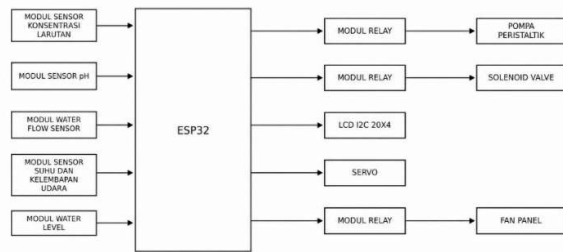


Figure 1. Block Diagram of the Hydroponic Nutrient Automation System Based on the ESP32 Microcontroller

The realization of the proposed circuit was conducted through the following steps:

1. Component Selection

The selection of components in this study was based on the requirements of a hydroponic nutrient automation system capable of monitoring nutrient solution quality, controlling water filling and drainage, adding nutrients, adjusting pH, circulating the nutrient solution, and displaying system conditions locally. The components used consist of a processing unit, sensors, a real-time clock module, display module, power supplies, a main switch, relay modules, and actuators.

The main components employed in this system include the ESP32-S3 DevKit C as the central controller, the DFRobot Gravity SEN0161-V2 pH sensor, the DFRobot Gravity Analog Electrical Conductivity Sensor/Meter v2 K=1 for EC/TDS measurement, the DFRobot DHT22 temperature and humidity sensor, the waterproof AJ-SR04M

ultrasonic sensor, the YS-S201 water flow sensor, the DS3231 real-time clock (RTC) module, a 20×4 LCD with an I2C module, a 5 V 4-channel 30 A dual-trigger relay module, INTLAB 12 V DC 2×4 NKP-DC-S10D peristaltic pumps, Mean Well LRS-50-5 and LRS-50-12 power supplies, a 1P ON/OFF rotary switch, 12 V DC solenoid valves, a Yamano WP-103 circulation pump, an MG996R servo motor, and a 12 V DC fan.

The selection of these components was carried out by considering their operating voltage compatibility, output signal types, microcontroller pin requirements, actuator capabilities, ease of integration, and system testing requirements. Analog sensors, such as the pH and EC/TDS sensors, are connected to the ADC inputs of the ESP32-S3, whereas digital sensors and communication modules, including the DHT22, AJ-SR04M, YS-S201, DS3231 RTC, I2C LCD, and servo motor, are connected according to their respective signal types. High-current actuators are not driven directly by the microcontroller pins but instead are controlled through relay modules or appropriate power sources.

2. Circuit Implementation

The implementation of the system circuitry is divided into five main sections: the control circuit, sensor circuit, display circuit, actuator circuit, and power supply circuit. The control circuit utilizes the ESP32-S3 DevKit C as the main controller. The analog pH and EC/TDS sensors are connected to the ADC pins of the ESP32-S3. The DHT22, AJ-SR04M, and YS-S201 sensors are connected to digital pins according to their respective reading requirements. The DS3231 RTC module and the 20×4 I2C LCD communicate through the SDA and SCL lines.

The actuator circuit employs a 12 V power source for the solenoid valves, peristaltic pumps, circulation pump, and DC fan. These loads are controlled using relays to prevent the load current from passing directly through the microcontroller pins. The MG996R servo motor is controlled using a PWM signal generated by the ESP32-S3 and is supplied with adequate power. The ground of the control circuit and the actuator signal ground are connected as a common reference, whereas the power lines for the loads are separated to prevent electrical noise generated by motors and solenoids from affecting sensor readings.

The Mean Well LRS-50-5 power supply provides a 5 V output for modules requiring a 5 V supply, while the Mean Well LRS-50-12 provides a 5 V output for modules requiring a 5 V supply, while the Mean Well LRS-50-12 provides a 12 V output for the actuators.

Table 1. Components Used in the Proposed System

No	Component Name	Function	Specifications	Formula
1.	ESP32-S3 Devkit C	Main controller for sensor reading, data processing, control algorithms, actuator control, and LCD communication.	Dual-core Xtensa LX7, 240 MHz, 3.3 V operation, ADC and I2C support, multiple GPIO pins.	
2.	DFRobot Gravity Analog pH Sensor/Meter Kit V2 (SEN0161-V2)	Measures the pH of the hydroponic nutrient solution.	Measurement range: pH 0–14, analog output, operating voltage: 3.3–5 V.	
3.	DFRobot Gravity Analog EC Sensor/Meter V2 (K=1)	Measures electrical conductivity and estimates nutrient concentration (TDS).	Measurement range: 0–20 mS/cm, analog output, calibration solutions: 1.413 and 12.88 mS/cm.	Electrical Conductivity (EC) to Total Dissolved Solids (TDS) (Thirumalini & Joseph, 2009): $TDS = K \times EC$ [29] where: TDS = Total Dissolved Solids (ppm or mg/L) K = EC-to-TDS conversion factor (typically 0.5–0.7 depending on the solution) EC = Electrical Conductivity (μ S/cm)
4.	DFRobot DHT22 Temperature and Humidity Sensor	Measures ambient temperature and relative humidity around the hydroponic system.	Temperature: –40 to 80°C, Humidity: 0–100% RH, digital output.	
5.	Waterproof Ultrasonic Sensor AJ-SR04M	Detects the nutrient tank water level.	Measurement range: 25–450 cm, waterproof probe, operating voltage: 5 V.	Water Level Calculation (Borankulova et al., 2025) If: T = Height of the tank D = Measured distance from the ultrasonic sensor to the water surface then $H = T - D$
6.	Water Flow Sensor YS-S201	Detects water flow as a safety interlock for dosing operations.	Operating voltage: 5–18 V, pulse output, flow rate: 1–30 L/min.	
7.	RTC DS3231 Module	Provides real-time clock information and schedules nutrient replacement.	I2C communication, integrated backup battery, accuracy ± 2 ppm.	
8.	LCD 2004 with I2C Module V3.0	Displays sensor readings and system status.	20 columns $\times$ 4 rows, I2C interface, operating voltage: 5 V.	
9.	Relay Module 5V 4CH 30A Dual Trigger	Controls high-current actuators such as pumps and valves.	4 channels, trigger modes: active HIGH/LOW, load capacity: 30 A.	
10.	INTLAB 12 V DC Peristaltic Pump (NKP-DC-S10D)	Doses nutrient solutions and pH adjustment liquids.	Operating voltage: 12 V DC, controlled flow based on activation time.	Volume Calculation (Mandal, 2009) $V = Q \times t$ where: V = Volume (L or mL) Q = Flow rate (L/s or mL/s) t = Time (s) Electrical Power (Luo, Ranzi, & Dong, 2024) $P = V \times I$ P = Electrical power (W) V = Voltage (V) I = Current (A)
11.	Mean Well LRS-50-5 and LRS-50-12 Power Supplies	Provide regulated 5 V and 12 V power for the system.	Output: 5 V/10 A and 12 V/4.2 A, rated power: 50 W.	
12.	Rotary Switch 1P ON/OFF	Main switch for turning the system on and off.	Single-pole manual rotary switch.	
13.	Solenoid Valve 12 V DC	Controls water inflow automatically.	Operating voltage: 12 V DC, normally closed type.	
14.	Yamano WP-103 Circulation Pump	Circulates the nutrient solution to ensure homogeneous mixing.	Operating voltage: 12 V DC, submersible water pump.	
15.	Servo MG996R	Opens and closes the drain valve mechanism.	Operating voltage: 4.8–7.2 V, torque up to 11 kg·cm, PWM control.	Servo Angle as a Function of PWM (Autso, Kudelina, Vaimann, Rassölkin, & Kallaste, 2024) $\Theta = f(\text{PWM})$ θ = Servo rotation angle (°) PWM = Pulse Width Modulation signal (μ s or duty cycle)
16.	DC Fan 12 V	Provides air circulation inside the control panel.	Operating voltage: 12 V DC, brushless fan.	

A 1P ON/OFF rotary switch is installed as the main system switch. With this power distribution arrangement, the system can operate more stably because the sensor loads and actuator loads are supplied through separate power lines.

A. Software Implementation

The system software was developed using the Arduino IDE platform and programmed in the C++ programming language. The program was designed to manage system initialization, sensor data acquisition, flow validation, pH and EC/TDS data processing, water-level control, nutrient dosing regulation, periodic nutrient solution replacement, RTC time acquisition, servo and fan control, and LCD display management. The program operates continuously in a repetitive loop, enabling the system to monitor the condition of the nutrient solution in real time.

B. Overall System Implementation

The overall system implementation integrates all components into a single hydroponic nutrient automation system. When the system is turned on using the rotary switch, the 5 V and 12 V power supplies provide electrical power to the control circuit and actuators. The ESP32-S3 initializes the sensors, RTC, LCD, relays, servo motor, and actuators. The system first reads the water level using the AJ-SR04M ultrasonic sensor. If the water level is insufficient, the filling solenoid valve is opened until the water reaches the predetermined level. Once the water level is adequate, the Yamano WP-103 circulation pump is activated, and the water flow is verified using the YS-S201 flow sensor. If normal flow is detected, the ESP32-S3 reads the pH and EC/TDS values to determine the requirements for nutrient addition or pH adjustment.

If the EC/TDS value is below the operating threshold, the nutrient A and nutrient B peristaltic pumps are activated gradually. If the pH value falls outside the specified range, the pH-up or pH-down pump is activated accordingly. The solenoid valves and servo motor are used for the filling and drainage processes, while the DS3231 RTC module determines the schedule for periodic nutrient solution replacement. All sensor readings and actuator statuses are displayed on the 20×4 I2C LCD. With this configuration, the system is expected to maintain the condition of the hydroponic nutrient solution automatically, reduce dependence on manual monitoring, and improve the safety of the dosing process through the implementation of a flow-based safety interlock. The realization of the overall system design is shown in Fig. 2.



Figure 2. Final Implementation of the Proposed System

The system operation flowchart describes the procedures followed by users when operating the automatic hydroponic nutrient system. The operation begins by turning on the system using the main switch. After activation, the ESP32-S3 initializes the sensors, LCD, RTC, relays, servo motor, and actuators connected to the system. Subsequently, the user can view the initial information displayed on the 20×4 I2C LCD. After the initialization process is completed, the system reads the water level in the nutrient solution tank. If the water level has not reached the predetermined threshold, the system activates the filling solenoid valve until the water reaches the normal level.

Once the water level is sufficient, the circulation pump is activated, and the water flow sensor detects the presence of water flow. If flow is detected, the system proceeds to measure the pH and EC/TDS values. However, if no flow is detected, the system suspends the dosing process and displays a warning message on the LCD.

If the EC/TDS value is below the specified threshold, the system gradually activates the nutrient A and nutrient B pumps. If the pH value is outside the desired range, the system activates either the pH-up or pH-down pump as required. In addition, the system reads the time from the DS3231 RTC module to determine the nutrient replacement schedule. When the scheduled nutrient replacement time is reached, the system opens the drainage valve using the servo motor and subsequently refills the tank through the solenoid valve.

III. RESULTS

Module testing was conducted to determine whether each component of the ESP32-based hydroponic nutrient automation system operates according to its intended function. The tests were performed gradually, beginning with the sensor modules, processing modules, actuator modules, display module, and finally the overall system. The modules tested consisted of the EC sensor, pH sensor,

ultrasonic sensor, temperature and humidity sensor, flow sensor, relay module, peristaltic pumps, solenoid valves, circulation pump, RTC module, 20×4 I2C LCD, power supply, and ESP32 microcontroller. Testing was carried out by comparing the readings obtained from the developed system with standard solutions or reference conditions. The difference values and percentage errors were used to determine the accuracy of the sensor readings. The percentage error was calculated using:

$$\text{Error (\%)} = \frac{|\text{Reference Value} - \text{Measured Value}|}{\text{Reference Value}} \times 100\%$$

Through these tests, it can be determined whether each module functions properly before evaluating the entire system.

A. EC Sensor Testing and Analysis

The EC sensor was tested to evaluate its ability to measure the concentration of hydroponic nutrient solutions. The EC sensor is one of the main sensors in the system because it serves as the basis for controlling the addition of nutrient A and nutrient B. The tests were conducted using standard solutions of 1.413 mS/cm and 12.88 mS/cm. The sensor readings were compared with the standard values to determine the measurement differences and percentage errors.

Based on Table 2, the EC sensor could measure two standard solutions with different concentrations, namely 1.413 mS/cm and 12.88 mS/cm. For the 1.413 mS/cm standard solution, the measurement deviation ranged from 0.073 mS/cm to 0.187 mS/cm, with the highest error reaching 13.23%. The largest error occurred during the third measurement, indicating that the sensor exhibited greater deviations at lower EC values than at higher EC values.

For the 12.88 mS/cm standard solution, the measurement deviation ranged from 0.28 mS/cm to 0.38 mS/cm, corresponding to errors between 2.17% and 2.95%. These results indicate that the sensor provides more stable readings at higher EC values. The average error obtained for the 12.88 mS/cm solution was lower than that obtained for the 1.413 mS/cm solution.

Overall, the EC sensor could detect changes in solution concentration from low to high values. Measurement deviations may be influenced by sensor voltage stability, solution temperature, electrode condition, sensor stabilization time, and calibration procedures. Despite some deviations, the test results indicate that the EC sensor is suitable for use as an input to determine whether the nutrient concentration is below, within, or above the specified range.

B. pH Sensor Testing and Analysis

The pH sensor was tested to evaluate its capability to measure the acidity level of the solution. The sensor serves as the basis for determining whether the system should add pH-up or pH-down solution. The tests were conducted using pH 4 and pH 7 buffer solutions.

Table 1. EC Sensor Calibration Results Using 1.413 mS/cm and 12.88 mS/cm Standard Solutions

Measurement No-	Reference Value (mS/cm)	Sensor Reading (mS/cm)	Difference (mS/cm)	Error (%)
1	1.413		0.073	5.16
2	1.413		0.143	10.12
3	1.413		0.187	13.23
4	12.88		0.38	2.95
5	12.88		0.32	2.48
6	12.88		0.28	2.17

Based on Table 3, the pH sensor successfully measured both buffer solutions with values close to the references. For the pH 4 buffer solution, the sensor readings ranged from 3.99 to 4.00, with a maximum deviation of 0.01 and a maximum error of 0.25%. For the pH 7 buffer solution, the readings ranged from 6.97 to 6.98, with a maximum deviation of 0.03 and a maximum error of 0.43%.

Although slight differences were observed between the reference values and sensor readings, the resulting errors remained relatively small. These differences may be caused by probe condition, solution temperature, sensor stabilization time, and calibration procedures.

Overall, the pH sensor demonstrated good performance because its readings were very close to the buffer solution values. Therefore, the sensor can be used as an input for determining whether the hydroponic solution requires pH adjustment to maintain suitable conditions for plant growth.

C. AJ-SR04M Ultrasonic Sensor Testing and Analysis

The ultrasonic sensor was tested to determine its ability to measure the water level in the nutrient tank. The sensor was mounted at the top of the tank and

measured the distance between the sensor and the water surface.

Table 2. pH Sensor Performance Test

Buffer Solution	Measurement No	Test Result	Sensor Reading	Difference	Error (%)
pH 4	1		3.99	0.01	0.25
pH 4	2		4.00	0	0
pH 4	3		3.99	0.01	0.25
pH 7	1		6.97	0.03	0.43
pH 7	2		6.97	0.03	0.43
pH 7	3		6.98	0.02	0.29

The water level was calculated using:

$$\text{Water Height} = \text{Tank Height} - \text{Distance from Sensor to Water Surface}$$

Based on Table 4, the ultrasonic sensor measured the water level accurately. In the first four measurements, the difference between the reference and measured values was 0 cm, resulting in an error of 0%. In the fifth measurement, a difference of 1 cm was observed, corresponding to an error of 5%.

Table 3. Ultrasonic Sensor Test Results

Measurement No.	Reference Water Level (cm)	Sensor Reading (cm)	Difference (cm)	Error (%)
1			0	0 ×
2			0	0
3			0	0
4			0	0
5			1	5

This deviation occurred because the AJ-SR04M ultrasonic sensor has a minimum sensing distance of approximately 20 cm. When the measured

distance falls below this limit, the sensor cannot accurately identify the actual distance.

Overall, the ultrasonic sensor can be used to determine whether the water level is low, normal, or full. When the water level is below the specified threshold, the system activates the filling solenoid valve. Once the water reaches the maximum level, the valve is turned off. Therefore, the ultrasonic sensor plays an important role in automatic water filling and overflow prevention.

D. DHT22 Temperature and Humidity Sensor Testing and Analysis

The temperature and humidity sensor was tested to evaluate its ability to measure environmental conditions around the plants. The measurements were compared with reference temperature and humidity instruments.

Based on Tables 5, the DHT22 sensor successfully measured the ambient temperature and humidity around the hydroponic system. Although the sensor is not used as the primary control parameter for nutrient management, it provides useful information regarding the environmental conditions of the plants. The measured temperature and humidity values are displayed on the LCD so that users can directly monitor the surrounding conditions.

Table 4. DHT22 Temperature and Humidity Sensor Test Results

Measurement No.	Temperature Sensor (°C)	Humidity Sensor (%)
1		
2		
3		

E. Flow Sensor Testing and Analysis

The flow sensor was tested to determine its ability to detect water flow in the circulation line. In this system, the flow sensor is not used to measure the solution volume but instead functions as a safety interlock. The test results showed that the sensor could successfully distinguish between flow and no-flow conditions. When no flow was detected, the system did not activate the nutrient pumps or pH adjustment pumps. This function is essential to prevent incorrect sensor readings and excessive nutrient addition. The implementation of this safety

interlock makes the system safer because the dosing process is only performed when the nutrient solution is circulating properly.

Table 6 shows the flow sensor test results. The results indicate that the flow sensor can reliably distinguish between flow and no-flow conditions. When no water flow is detected, the system prevents the activation of both the nutrient dosing pumps and the pH adjustment pumps. This mechanism is essential to avoid inaccurate sensor readings and prevent excessive dosing of nutrient or pH adjustment solutions. By implementing this safety interlock, the system ensures that the dosing process is performed only when the nutrient solution is circulating properly, thereby improving the reliability and operational safety of the hydroponic nutrient automation system.

Table 5. Flow Sensor Test Results

Test Condition	Flow Condition	Flow Sensor Output	Result
Circulation pump OFF	No flow		Pass
Circulation pump ON	Flow detected		Pass

F. Relay Module and Actuator Testing and Analysis

The relay module and actuators were tested to ensure that the ESP32 could properly control all output devices. The tested actuators included the nutrient A pump, nutrient B pump, pH-up pump, pH-down pump, filling solenoid valve, drainage solenoid valve, and circulation pump.

Based on Table 7, the relay module successfully controlled all actuators according to the commands generated by the ESP32. The relay acts as an electronic switch that isolates the low-voltage control circuit from actuators operating at different voltage levels. These results indicate that the actuator system performs as required for hydroponic automation processes.

Table 6. Actuator Functional Test Results

Actuator	ESP32 Command	Expected Condition	Test Result	Result
Nutrient Pump A	ON	Pump ON	Pump ON	Pass
Nutrient Pump A	OFF	Pump OFF	Pump OFF	Pass
Nutrient Pump B	ON	Pump ON	Pump ON	Pass
Nutrient Pump B	OFF	Pump OFF	Pump OFF	Pass
pH Up Pump	ON	Pump ON	Pump ON	Pass
pH Up Pump	OFF	Pump OFF	Pump OFF	Pass
pH Down Pump	ON	Pump ON	Pump ON	Pass
pH Down Pump	OFF	Pump OFF	Pump OFF	Pass
Filling Solenoid Valve	ON	Valve OPEN	Valve OPEN	Pass
Filling Solenoid Valve	OFF	Valve CLOSED	Valve CLOSED	Pass
Circulation Pump	ON	Pump ON	Pump ON	Pass
Circulation Pump	OFF	Pump OFF	Pump OFF	Pass

G. Peristaltic Pump Flow Rate Testing

The peristaltic pump flow rate test was conducted to determine the amount of solution delivered within a certain period. This test is important because nutrient and pH solution additions are controlled based on pump operating time.

The pump flow rate was calculated using:

$$Q \text{ (Pump Flow Rate)} = \frac{V \text{ (Solution volume)}}{t \text{ (pump operating time)}}$$

Based on Table 8, nutrient A, nutrient B, and pH-up pumps required 110 seconds to deliver 100 mL of solution, resulting in a flow rate of 0.91 mL/s. Meanwhile, the pH-down pump required 111 seconds to reach the same volume, corresponding to a flow rate of 0.901 mL/s. The flow rate difference among the pumps was only approximately 0.009 mL/s, indicating that all peristaltic pumps exhibited nearly identical performance. The average flow rate of the four pumps was approximately 0.908 mL/s. The designed hydroponic system uses 20 L of water. To achieve a nutrient concentration of approximately 1000–1300 ppm, a nutrient-to-water ratio of 1:1:200

is applied. Based on this ratio, 20 L of water requires 100 mL of nutrient A and 100 mL of nutrient B. With a pump flow rate of approximately 0.91 mL/s, the required pump operating time is about 110 seconds. Therefore, the operating duration of the nutrient pumps can be used as the basis for the initial nutrient dosing process.

Table 7. Peristaltic Pump Flow Rate Test Results

Pump	Target Volume (mL)	Time Required (s)	Flow Rate (mL/s)
Nutrient A	100	110	0.91
Nutrient B	100	110	0.91
pH Up	100	110	0.91
pH Down	100	111	0.901

H. RTC DS3231 Testing and Time Synchronization

The RTC module was tested to ensure that the DS3231 could store and display time in real time. The RTC is used to execute scheduled operations, including automatic nutrient solution replacement every two weeks.

Based on Table 9, the RTC displayed both date and time correctly. The module was also able to retain the time information even when the system was powered off because it is equipped with a backup battery. Therefore, the RTC can be reliably used as the time reference for scheduled automation processes.

Table 8. RTC DS3231 Performance Test

Reference Time	RTC Time	Difference	Result
16:50 21/05/2026	16:50 21/05/2026	0	Pass



I. 20×4 I2C LCD Testing

The 20×4 I2C LCD was tested to determine whether it could display the information required by the user. The displayed information includes EC/TDS values, pH values, temperature, humidity, date, time, and system status. Based on Table 10, the LCD successfully displayed all system information in real time. The use of the I2C interface reduces the number of microcontroller pins required, making the display easier to integrate with the ESP32-based system. The LCD enables users to monitor system conditions without requiring a computer or internet connection.

Table 10. 20×4 I2C LCD Performance Test

Test No.	Displayed Data	Result
1	EC/TDS and pH	Pass
2	Temperature and Humidity	Pass
3	Date and Time	Pass
4	System Status	Pass

J. Overall System Testing and Analysis

The overall system testing was conducted to determine whether all modules could operate in an integrated manner according to the designed sequence. The test procedure included water filling, circulation pump activation, flow detection, EC/TDS and pH sensor readings, nutrient dosing, pH adjustment, LCD display operation, and automatic nutrient replacement.

Based on table 11, the system successfully performed the hydroponic automation process step by step. When the water level was low, the system activated the filling solenoid valve until the water reached the specified level. After sufficient water was available, the circulation pump operated and the flow sensor verified the presence of water flow. If no flow was detected, the system did not activate the dosing process as part of the safety interlock mechanism.

During nutrient control testing, the system successfully read the EC/TDS values and activated the nutrient pumps when the nutrient concentration was below the setpoint. After nutrient addition, the solution was circulated to ensure proper mixing before subsequent measurements were taken.

During pH control testing, the system correctly identified low, normal, and high pH conditions and activated the pH-up or pH-down pumps as required. All system statuses were displayed on the 20×4 I2C LCD, allowing users to monitor the system directly.

Table 11. Overall System Performance Test

Test Stage	Input Condition	Expected Response	Result
Initial cycle reset	Drain water	Drain servo is activated	Cycle reset completed; drain servo activated; water level reached 12.7 cm
Water filling	Low water level	Filling solenoid valve is activated	Initial filling cycle performed; water level increased from 0.0 cm
Full tank	Water level reaches the upper limit	Filling solenoid	Tank reached full level; filling

Test Stage	Input Condition	Expected Response	Result
		valve is deactivated	solenoid valve deactivated
Nutrient solution circulation	Circulation pump activated	All sensors are read	Circulation pump operated normally; all sensors were successfully read
Safety interlock	No flow detected	Error message is displayed	Error message displayed successfully
Low TDS	TDS value below the setpoint	Nutrient Pumps A and B are activated	Nutrient Pumps A and B activated successfully
Normal TDS	TDS value at the setpoint	Nutrient pumps are deactivated	Nutrient pumps deactivated; water circulation remained active
High TDS	TDS value above the setpoint	Filling solenoid valve is activated	Filling solenoid valve activated successfully
Low pH	pH below the lower limit	pH Up pump is activated	pH Up pump activated for 5 s
High pH	pH above the upper limit	pH Down pump is activated	pH Down pump activated for 5 s
Normal pH	pH within the acceptable range	pH adjustment pumps are deactivated	pH adjustment pumps are deactivated

IV. DISCUSSION

Based on both module-level and overall system testing, the ESP32-based hydroponic nutrient automation system successfully performed all its intended functions. The EC/TDS and pH sensors served as the primary inputs for nutrient solution quality control. The ultrasonic sensor measured the water level in the tank, while the flow sensor acted as

a safety mechanism to ensure that dosing occurred only when the solution was circulating properly. The temperature and humidity sensor provided additional information regarding environmental conditions.

From the actuator perspective, the peristaltic pumps successfully delivered nutrients and pH solutions in a controlled manner, while the solenoid valves effectively performed water filling and drainage processes. The relay module reliably transmitted control signals from the ESP32 to the actuators. The 20×4 I2C LCD successfully displayed sensor data and system status information, facilitating system monitoring.

The safety interlock feature represents one of the most important aspects of the proposed system. Without the flow sensor, the system could potentially add nutrients when the solution is not circulating, causing sensor readings that do not represent the actual condition of the nutrient solution. By incorporating the flow sensor, the system performs measurements and dosing only when flow is detected. This approach makes the system safer and reduces the risk of nutrient and pH solution overdosing.

Overall, the proposed system is capable of automatically performing water filling, nutrient monitoring, pH adjustment, water-level detection, flow detection, and periodic nutrient replacement. The system reduces dependence on manual supervision and helps maintain nutrient solution stability in small- to medium-scale hydroponic systems. If it is practicable, the research findings are discussed and contrasted with the findings from the papers cited.

V. CONCLUSION

Based on the design, implementation, and experimental testing, the ESP32-S3-based hydroponic nutrient management system was successfully developed and demonstrated its intended functions under the tested laboratory conditions. The prototype performed automated water filling, circulation verification, EC/TDS and pH measurement, nutrient dosing, pH adjustment, and programmed nutrient solution replacement. The pH sensor achieved a maximum measurement error of 0.43%, while the EC sensor showed errors of 5.16–13.23% at 1.413 mS/cm and 2.17–2.95% at 12.88 mS/cm, indicating greater measurement deviation at the lower EC level. The AJ-SR04M ultrasonic sensor successfully detected water-level conditions, with errors of 0% in four measurements and 5% in one measurement. The YS-S201 flow sensor successfully distinguished flow and no-flow conditions and prevented nutrient and pH dosing when circulation was absent. The relay module and actuators operated

according to the programmed control logic, while the four peristaltic pumps achieved an average flow rate of approximately 0.908 mL/s, providing a reference for pump-based dosing control. The integrated system successfully completed the tested automation sequence, including water filling, circulation, flow verification, nutrient control, pH adjustment, and scheduled solution replacement. However, the evaluation was limited to laboratory-based functional testing with a limited number of calibration points and short test duration. Long-term nutrient stability, repeated dosing accuracy, and plant-growth performance were not evaluated. Therefore, the results demonstrate the functional feasibility of the prototype under the tested conditions, while further long-term testing is required to evaluate its performance in actual hydroponic cultivation.

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